MILITARY SPECIFICATION

ELECTRONIC ASSEMBLY FOR MINE, ANTIPERSONNEL: HE, M74

This specification is approved for use by the U.S. Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the detail requirements not normally covered by the drawings and provides quality assurance provisions for the fabrication and inspection of an electronic assembly used in the Mine, Antipersonnel: HE, M74.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-E-17555 - Electronic and Electrical Equipment Accessories, and Repair Parts, Packaging and Packing of

MIL-S-45743 - Soldering, Manual Type, High Reliability Electrical and Electronic Equipment

FSC: 1345

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-s-46844 - Solder Bath Soldering of Printed Wiring Assemblies

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

MIL-P-50884 - Printed Wiring, Flexible General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes

MIL-STD-202 - Test Methods For Electronic and Electrical Component Parts

MIL-STD-750 - Test Methods For Semiconductor Devices

MIL-STD-883 - Test Methods and Procedures For Microelectronics

MIL-STD-1188 - Commerical Packing of Supplies and Equipment

DRAWINGS (See 6.4)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

PRODUCT DRAWINGS

9292960 - Electronic Assembly Timing Operations, Oscillator Periods and Low Voltage Detection Levels(U)

9292965 - Electronic Assembly

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specification procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings, specifications and standards.

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3.2 Electronic assembly.

3.2.1 Parts. Electronic assembly parts shall comply with the requirements specified on Drawing 9292965, associated drawings, and applicable specifications and standards.

3.2.2 Operation. Electronic assembly timing, signal generation, and low voltage detect operations specified in 3.4.2 shall be in accordance with the requirements on Drawing 9292960.

3.3 Soldering. Soldering procedures and acceptance criteria for the soldered components shall be in accordance with MIL-S-45743 and MIL-S-46844, except as specified in Appendix herein.

3.3.1 Solderability. Electronic assembly parts that are to be soldered shall comply with the solderability requirements of MIL-S-45743 and MIL-S-46844,

3.3.2 Quality assurance soldering plan. The contractor will generate a Quality Assurance Soldering Plan that will detail the overall soldering procedure. This plan shall include detailed information for handling, cleaning, hand soldering, machine soldering, touch up or rework, process controls tool maintenance, flagging procedures for avoiding excessive defects and all other pertinent QA procedures relative to soldering. The plan shall include the documentation requirements of MIL-S-46844. This plan will be generated and submitted to ARRADCOM, DRDAR-QAA-M (see 4.5.19, 6.5).

3.4 Performance characteristics.

3.4.1 Quiescent current. Electronic assembly quiescent current shall be less than 220 microampers when the supply voltage is 6.5 + 0.10 volts, the ambient temperature 25° ± 5°C and the input/output terminals loaded as specified in 4.5.1.

3.4.2 Timing, signal, and detection operations.

3.4.2.1 Primary Time Base (PTB) oscillator period. The PTB oscillator period shall comply with 3.2.2 when tested as specified in 4.5.2.

3.4.2.2 Test Time Base (TTB) oscillator period. The TTB oscillator period shall comply with 3.2.2 when tested as specified in 4.5.3.

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3.4.2.3 S&A assembly arm signal. The PTB clock count at which the Safety and Arming (S&A) assembly arm signal turns on the piston actuator firing circuit shall comply with 3.2.2 when tested as specified in 4.5.4.

3.4.2.4 Pressure cartridge signal. The PTB clock count at which the pressure cartridge signal turns on the pressure cartridge firing circuit shall comply with 3.2.2 when tested as specified in 4.5.5.

3.4.2.5 Electronic arm. The PTB clock count at which the electronic assembly is electronically armed shall comply with paragraph 3.2.2. The electronic assembly shall not process destruct signals from the low voltage detect, tripline-sensor interface and antidisturbance circuits until after electronic arm. The PTB clock count that initiates electronic arm and the ability of the electronic assembly to inhibit destruct signals prior to electronic arm shall be tested as specified in 4.5.6.

3.4.2.6 Low voltage detect. The supply voltage level at which the low voltage detect circuit turns on the detonator firing circuit shall comply with 3.2.2 when tested as specified in 4.5.7.

3.4.2.7 Extended self-destruct signal. The PTE clock count at which the extended self-destruct signal turns on the detonator firing circuit shall comply with 3.2.2 when tested as specified in 4.5.8.

3.4.2.8 Normal self-destruct signal. The PTB clock count at which the normal self-destruct signal turns on the detonator firing circuit shall comply with 3.2.2 when tested as specified in 4.5.9.

3.4.3 Target initiated destruct modes. In compliance with 3.4.2.5 the requirements specified in 3.4.3.1 and 3.4.3.2 apply after electronic arm.

3.4.3.1 Antidisturbance, The antidisturbance circuit shall initiate the turn on of the detonator firing circuit when the antidisturbance switch changes state; that is, a transition from either an open or closed switch condition. This requirement shall be tested as specified in 4.5.10.

3.4.3.2 Tripline. The tripline-sensor interface circuit shall initiate the turn on of the detonator firing circuit for the ‘ conditions specified in 3.4.3.2.a; however, it shall not initiate the turn on of the detonator firing circuit for the conditions specified in 3.4.3.2.b. The sensitivity of the tripline-sensor interface circuit shall be tested for as specified in 4.5.11.

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a. Turn on conditions. A resistance change from 3.9 x 10^3 ohms + 5% to 10 x 10^3 ohms + 5% that occurs after electronic arm at any one of the four inputs of the tripline-sensor interface circuit shall cause the turn on of the detonator firing circuit.

b. Non-turn on conditions. A resistance change from a short (<10 ohms) to 10K ohms (± 5%) at each input of the tripline sensor interface circuit before electronic arm shall not cause the turn-on of the detonator firing circuit at or after electronic arm. A resistance change from a short (< 10 ohms) to 3.9K ohms (± 5%) at each input of the tripline sensor interface circuit after electronic arm shall not cause the turn-on of the detonator firing circuit.

3.4.4 Primary Time Base (PTB) oscillator check. The oscillator check circuit shall initiate the turn on of the detonator firing circuit when the PTB oscillator period becomes greater than the test time base oscillator period. This requirement shall be tested as specified in 4.5.12.

3.4.5 Detonator fire signal output energy. The electonic assembly shall provide energy to fire the piston actuator, or the pressure cartridge, or the detonator when required as described in this specification. The output energy shall be such that a minimum of three (3) volts shall be measured no more than forty (40) microseconds after a fire output signal is generated, and a minimum of two (2) volts is measured at a minimum time of one hundered twenty (120) microseconds after the fire output signal is generated. Voltages shall be measured at the applicable output terminals across a 3.9 ohm (± 5%) resistive load. This requirement shall be tested as specified in 4.5.13.

3.4.5.1 Piston actuator firing circuit. When turned on, the piston actuator firing circuit output signal shall remain on for only one PTB clock counts, that is, a time interval equal to one PTB oscillator period.

3.4.5.2 Pressure cartridge firing circuit. When turned on, the pressure cartridge firing circuit output signal shall remain on for only one PTB clock count.

3.4.5.3 Detonator firing circuit. When turned on, the detonator firing circuit output signal shall remain on for as long as the electronic assembly is functionally operational or is intentionally reset.

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3.4.6 Self-sterilization. Within 5 seconds after the turn on of the detonator firing circuit the voltage on capacitor C2 shall be less than 1.3 volts. It shall remain less than 1.3 volts for as long as the detonator firing circuit remains on. This requirement shall be tested as specified in 4.5.14.

3.4.7 Reset circuit. When a linear increasing voltage (zero to 6.5 + 0.10 volts in 50 milliseconds rein) is applied to the electronic assembly from an external d.c. power supply, the reset circuit (C4, R9-U-15) shall reset the IC's 9292962, 9292963. This requirement shall be tested as specified in 4.5,15.

3.5. High temperature operation. The electronic assembly shall meet the performance characteristics specified in 3.4 at 630 +3°C when tested as specified in 4.5.16.

3.6 Low temperature operation. The electronic assembly shall meet the performance characteristics specified in 3.4 at minus 370 -3°C when tested as specified in 4.5.16.

3.7 Inspection data. Data shall be recorded and retained for all First Article and Quality Conformance tests required by this specification in accordance with 4.5.20.

3.7.1 Variables data. The Primary Time Base Oscillator period, Test Time Base Oscillator period, Quiescent Current, and Low Voltage Detect Circuit Trigger Level shall be measured and recorded as variable data for First Article (see 4.5.21).

3.8 First article inspection. This specification contains provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. The electronic assembly shall be fabricated and finished in a thorough, workmanlike manner. It shall be free of burrs, chips, sharp edges, cracks, unblended radii, surface defects, dirt, grease, rust, corrosion and other foreign matter that may impair proper functioning. Components shall not be injured by the cleaning method, nor contaminated by the cleaning agent. All markings made during and after assembly, and markings on components when practical, shall be displayed and visibly defined. The electronic assembly shall be manufactured and/or processed in such a manner as to be uniform in quality and free from defects that will affect life, configuration and serviceability.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein, or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification,

4.2 Classification of inspections. The following types of inspections shall be conducted on this item.

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

Part Description	Part Identifier	Quantity
Support, Flexible, Printed Wiring	9292971	10
Electronics Assembly	9292965	24

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection See MIL-A-48078.

4.3.4 Requalification. The electronic assembly shall be subjected to another first article as stipulated in the contract if:

a. The contractor's maunfacturing techniques or process controls which affect the form, fit, or function of the assembly are revised by either the contractor or one of his sub-contracor supplying parts or assemblies.

b. The contractor relocated his assembly facilities.

c. The contractor uses a new sub-contractor for material, parts or assembly.

d. A change occurs in the design, material, performance or assembly.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Electronic Assembly				See Below
					NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Support, Flexible, Printed Wiring (P/N 9292971)	10			
	Examination for defects			3.2	4.4.2.3, MIL-P-50884
	Solderability			3.3.1	APPENDIX
	Electronic Assembly (Dwg. 9292965)				
	Mechanical	24		3.2	4.4.2.4
	Solder workmanship	All		3.3	4.4.2.4
	Performance Characteristics @ +25° ± 5°C	8		3.4	4.4.2.5, 4.5.16
	Performance Characteristics @ -37° ± 3°C	8		3.4,3.6	4.4.2.5, 4.5.16
	Performance Characteristics @ +63° ± 3°C	8		3.4,3.5	4.4.2.5, 4.5.16

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots shall contain Electronic Assemblies where each of the following listed items shall be identified by its drawing number (as a minimum, when space is not available, the last three digits) plus same revision letter, from not more than three lots from the same manufacturer:

Integrated circuit, CMOS, quad comparator.

Integrated circuit, special purpose NOR gate.

Integrated circuit, CMOS, timers.

Networks, resistor, SIP/DIP package.

Fuses.

Wiring, Printed, Flexible,

4.4.2 Examination, (See MIL-A-48078).

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Tests Forms, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, general inspection level II.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Stiffener				9292969
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR /100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Diameter of through hole for lead retainer boss, 4 places		0.40%	3.2	Gage
104	Location of C.D. #103 through hole, 4 places		0.40%	3.2	Gage
105	Profile of cutout for selector switch		0.40%	3.2	Gage
Minor					
201	Hole dimensions and locations		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Wiring, Printed, Flexible				9292970
					NEXT HIGHER ASSEMBLY
					9292971
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Visual and dimensional examination		0.40%	3.2	MIL-P-50884, 1/
102	Slit edge less than 0.01 inch from active conductor pattern		0.40%	3.2	Visual/Gage, 2/
103	Continuity missing between Hole No. 235 and Holes 291 and 288		100%	3.2	Resistance Tester, 3/
104	Continuity missing between Hole No. 230 and Hole No. 182		100%	3.2	Resistance Tester, 3/
105	Continuity missing between Hole No. 282 and Hole No. 233		100%	3.2	Resistance Tester, 3/
106	Continuity missing between Hole No. 278 and Hole No. 287		100%	3.2	Resistance Tester, 3/
107	Continuity missing between Hole No. 278 and Hole No. 248		100%	3.2	Resistance Tester, 3/
108	Continuity missing between Hole No. 292 and Hole No. 228		100%	3.2	Resistance Tester, 3/
109	Solderability		0.40%	3.3.1	4.5.19, 1/
Minor					
201	Flexible printed wire contamination		0.65%	3.2	Visual, 4.5.17

NOTES: 1/ See MIL-P-50884, Methods of Examination and Test.
2/ All approved measuring device shall be used when any apparent defect is found during visual inspection.
3/ Test current limited to 10 milliamperes.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Support, Flexible, Printed Wiring				9292971
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any missing or improperly mounted terminals		0.40%	3.2	Visual
102	Adhesive in unwanted noted area of dwg		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.4	Electronic Assembly				9292965
					NEXT HIGHER ASSEMBLY
					9292679
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Handling procedure - see 4.5.19				
Critical					
1	One or more fuse open		100%	3.2	Tester
2	Blocking or zener diode has improper polarity orientation; breakover or blocking voltage; open/short circuit (SD circuit)		100%	3.2	Tester
Major					
101	Conformal coat missing forward of noted area and components or in unwanted areas		0.40%	3.2	Visual (ultra-violet light)
102	Any component damaged, or missing, or improperly assembled	1/	0.40%	3.2	Visual
103	Incorrect orientation of the capacitor(s) polarity mark(s)	1/	0.40%	3.2	Visual
104	Switch Assembly height from board over max		0.40%	3.2	Gage
105	Component height from board over max, 2/		0.40%	3.2	Gage
106	Solder workmanship		3/	3.3	4.5.17, 4.5.19, APPENDIX
Minor					
201	Height of leads/solder on bottom of board over max		0.65%	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Before soldering. Other inspections refers to post soldering inspections. 2/ Applies to all components except switch assembly. 3/ See section 50 of the APPENDIX for sampling and acceptance criteria.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.5	Electronics Assembly (Non-destructive Tests)				9292965
					NEXT HIGHER ASSEMBLY 9292679
CATEGORY	EXAMINATION OR TEST Handling procedure - see 4.5.19	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Primary Time Base (PTB) oscillator period, under min. or over max.		100%	3.4.2.1	4.5.2
2	PTB clock count at extended self-destruct signal initiation, over max.		100%	3.4.2.7	4.5.8
3	PTB clock count at normal self-destruct signal initiation, over max.		100%	3.4.2.8	4.5.9
4	PTB clock count at S&A assembly arm signal initiation, under min.		100%	3.4.2.3	4.5.4
Special*	(* see 6.7)				
a	Test Time Base (TTB) oscillator period, over max.		100%	3.4.2.2	4.5.3
b	Oscillator check circuit does not detect a PTB period that is greater than the TTB period		100%	3.4.4	4.5.12
c	Low voltage detect circuit trigger level incorrect		100%	3.4.2.6	4.5.7
d	Self-sterilization circuit function incorrect		100%	3.4.6	4.5.14
Major					
101	PTB clock count at S&A assy arm signal over max.		4.4.3.1	3.4.2.3	4.5.4
102	Quiescent current over max.		4.4.3.1	3.4.1	4.5.1
103	PTB clock count at electronic arm incorrect		4.4.3.1	3.4.2.5	4.5.6

NOTES: 1/ All units shall be tested, at ambient, for functional parameters listed under critical and special.
2/ Except as noted in 4.4.3.1, a sample of the units shall be tested for all the functional parameters listed under critical, special and major at ambient, hot and cold. (see 4.5.16).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
4.4.2.5 Cont'd	Electronics Assembly (Non-destructive Tests)				9292965
					NEXT HIGHER ASSEMBLY 9292679
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
104	Sensor enable test - low voltage detect, tripline-sensor and anti-disturbance destruct signals not inhibited before electronic arm		4.4.3.1	3.4.2.5	4.5.6
105	TTB oscillator period under min.		4.4.3.1	3.4.2.2	4.5.3
106	PTB clock count at pressure cartridge signal incorrect		4.4.3.1	3.4.2.4	4.5.5
107	Antidisturbance switch mode function incorrect		4.4.3.1	3.4.3.1	4.5.10
108	Tripline-sensor interface circuit sensitivity test (test all four inputs)		4.4.3.1	3.4.3.2	4.5.11
109	Piston actuator firing circuit energy output incorrect		4.4.3.1	3.4.5.1	4.5.13
110	Pressure cartridge firing circuit energy output incorrect		4.4.3.1	3.4.5.2	4.5.13
111	Detonator firing circuit energy output incorrect		4.4.3.1	3.4.5.3	4.5.13
112	PTB clock count at extended self-destruct signal, under min.		4.4.3.1	3.4.2.7	4.5.8
113	PTB clock count at normal self-destruct signal, under min.		4.4.3.1	3.4.2.8	4.5.9
114	Reset circuit function incorrect		4.4.3.1	3.4.7	4.5.15
Minor 201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 3. Report all critical defects per 4.5.20.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Electronic Assembly				9292965
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Packaging incorrect		1/	5.1	MIL-E-17555
102	Packing incorrect		1/	5.2	MIL-E-17555
103	Marking incorrect		1/	5.3	MIL-E-17555, MIL-STD-1188
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Sample and accept-reject as per MIL-E-17555 for visual inspection.

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MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Capacitor, Fixed, Electrolytic-Tantalum, Solid (P/N 9287126-2, 82 uF)				9287126
					NEXT HIGHER ASSEMBLY 9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current over max		0.40%	3.2	1/
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Weld strength (two sides), 1/	8	2/	3.2	Pull Tester
105	Internal workmanship	20	3/	3.2	DWG 9292220
106	Breakdown voltage	10	3/	3.2	DWG 9292220
107	Sleeve missing		0.40%	3.2	Visual
108	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ See applicable note of dwg.
2/ Reject for one (1) or more defective.
3/ Reject for two (2) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.8	Capacitor, Fixed, Electrolytic-Tantalum, Solid (P/N 9287096-4, 39 uF)				9287096
					NEXT HIGHER ASSEMBLY 9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Leakage current over max		0.40%	3.2	1/
103	Weld strength (two sides), 1/	8	2/	3.2	Pull Tester
104	Internal workmanship	20	3/	3.2	DWG 9292220
105	Breakdown voltage	10	3/	3.2	DWG 9292220
106	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
107	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
108	Sleeve missing		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ See applicable note of dwg.
2/ Reject for one (1) or more defective.
3/ Reject for two (2) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Capacitor, Fixed, Plastic Dielectric (0.22 uF)				9287104
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Insulation resistance, 1/		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability (2 leads)	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ See applicable note of dwg.
2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Capacitor, Fixed, Ceramic, Dielectric (P/N M39014/02-1415, 1 uF)				MIL-C-39014/2
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Insulation resistance		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability (2 leads)	10	1/	3.3	MIL-STD-202, Method 208
105	Compliance with MIL-C-39014/2			3.2	Certification
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Capacitor, Fixed, Ceramic, Dielectric (P/N M39014/05-2037, 1000 pF)				MIL-C-39014/5
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Capacitance		0.40%	3.2	MIL-STD-202, Method 305
102	Insulation resistance		0.40%	3.2	MIL-STD-202, Method 302
103	Dissipation factor		0.40%	3.2	MIL-STD-202, Method 306
104	Solderability (2 leads)	10	1/	3.3	MIL-STD-202, Method 208
105	Compliance with MIL-C-39014/5			3.2	Certification
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.12	Semiconductor Device, Diode, Type JAN IN645-1				MIL-S-19500/240
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Reverse leakage current over max	1/	0.40%	3.2	MIL-STD-750, Method 4016d
102	Forward voltage		0.40%	3.2	MIL-STD-750, Method 4011
103	Solderability	10	2/	3.2	MIL-STD-202, Method 208
104	Evidence of JAN marking	10	2/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Color band end shall be used as cathode end during reverse current test, reverse current test is used to confirm that the color band is on cathode end of diode. 2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Semiconductor Device, Diode (P/N JAN IN750A-1)				MIL-S-19500/127
					NEXT HIGHER ASSEMBLY 9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Internal visual inspection			3.2	Certification
102	Reverse leakage current over max - cathode band at wrong end or missing 1/		0.40%	3.2	MIL-STD-750, Method 4016dc
103	Forward voltage drop		0.40%	3.2	MIL-STD-750, Method 4011
104	Breakdown voltage		0.40%	3.2	MIL-STD-750, Method 4022
105	Solderability, 2 leads	10	2/	3.2	MIL-STD-750, Method 2026
106	Compliance with MIL-S-19500/127			3.2	Certification
Minor					
201	Idenfication and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Color band end shall be used as cathode end during reverse current test, reverse current test is used to confirm that the color band is on cathode end of diode. 2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Fuse Electric				9328688
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bridgewire material			3.2	Certification
102	Fuse resistance		100%	3.2	Milliohmmeter, 3/
103	Case to lead resistance	1/	4/	3.2	Megohmmeter
104	No fire current for 10 second period, min.	1/,2/	4/	3.2	Tester
105	All fire energy (capacitor discharge) test	1/	4/	3.2	Tester
106	Solderability, 2 leads	10	4/	3.3	MIL-S-45743
Minor					
201	Fuse dimensioning	5	4/	3.2	Gage
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Sample size per MIL-STD-105, special inspection level S-4. 2/ Use these units for All Fire Energy Test (Major 105). 3/ Limit current to 1 ma. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Resistor, Fixed Composition, 100 K ohms, ± 5%, 1/4 W				MIL-R-39008/1 P/N RCR07G104JS 1/
					NEXT HIGHER ASSEMBLY 9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Resistance		0.40%	3.2	MIL-R-39008/1
102	Solderability, 2 leads	10	2/	3.2	MIL-STD-202, Method 208
Minor					
201	Identification and marking		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES:
1/ OPTIONAL: P/N RCR07G104JM.
2/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Switch Assembly				9298512
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Switch function seven (7) contact closures minimum each direction	1,2/	2.5%	3.2	4.5.22.1, 3/
102	Contact closure duration 0.60 seconds max.	1,2/	2.5%	3.2	4.5.22.1, 3/
103	Insulation resistance	1/,2/	2.5%	3.2	4.5.22.2, 3/
104	Assembly damaged, distorted or incomplete		0.40%	3.2	Visual
105	Ribbon improperly welded to can		0.40%	3.2	Visual
106	Epoxy seal missing or incomplete		0.40%	3.2	Visual
107	Plating of AD switch can	5/ BATCH		3.2	4.5.22.3
108	Solderability, 2 leads	10	4/	3.3.1	MIL-S-45743
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
	1/ All inspection lots of AD switches shall be subjected to 100% screening for proper function (Majors 101, 102 and 103) with all defectives removed. In addition to the screening operation, proper function shall be verified using the AQL's specified above.				
	2/ A defect is defined as: a switch with less than 7 momentary contact closures in each direction while being rotated 360° clockwise and counter clockwise; contact closure duration exceeding 0.60 second maximum; a switch with an insulation resistance, between switch terminals, of less than two (2) megohms. If any switch has any closure that exceeds 1.20 seconds the lot shall be rejected. The lot shall also be rejected if any switch has more than two contact closures between 0.60 and 1.20 seconds.				
	3/ See also drawing for test schematic and functional requirements.				
	4/ Reject for one (1) or more defectives.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9287122
4.4.2.17	Transistor (P/N 9287122-1, JAN 2N2907A, MIL-S-19500/291)				NEXT HIGHER ASSEMBLY 9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire material (See Note 2 of dwg)	32	1/	3.2	Visual
102	Wire bond (See Note 2 of dwg)	32	1/	3.2	Pull Tester, 2/
103	Compliance with MIL-S-19500/291			3.2	3/
104	Solderability (3 leads)	10	1/	3.2	MIL-S-45743
105	Leakage current Icer over max		0.40%	3.2	4/
106	Internal visual (pre-cap) see also note 9 of dwg.	32	1/	3.2	MIL-STD-750, Method 2072
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective. 2/ Pull all wires on each device. 3/ Visual evidence of JAN marking. 4/ See applicable note and test schematic of dwg.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9292966 / NEXT HIGHER ASSEMBLY 9292965
4.4.2.18	Resistor Network, Fixed, Film				
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR /100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability, all leads	10	1/	3.3.1	MIL-STD-202, Method 208
102	Network characteristics		2/	3.2	MIL-R-63322
103	Approved part/vendor verification	5	1/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective.
2/ Certification of conformance required (see 6.2).

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Resistor Network, Fixed, Film (Externally Trimmable)				9292942
					NEXT HIGHER ASSEMBLY 9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Solderability, all leads	10	1/	3.3.1	MIL-STD-202, Method 208
102	Network characteristics		2/	3.2	MIL-R-63319
103	Approved part/vendor verification	5	1/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual

NOTES: 1/ Reject for one (1) or more defective.
2/ Certification of conformance required (see 6.2).

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.21	Microcircuit, Digital, CMOS, Special Purpose NOR Gate	1 OF 1	9292968
			NEXT HIGHER ASSEMBLY 9292965

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire Bond strength	5	1/	3.2	Pull Tester
102	Wire Material	5	3/	3.2	Visual
103	Solderability	5	3/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics		2/	3.2	MIL-M-63324
105	Screening		2/	3.2	MIL-M-63324
106	External visual		2/	3.2	MIL-M-63324
107	Approved part/vendor verification	5	3/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
	Item requires special handling - See 4.5.18				

NOTES:
1/ Pull all wires on each device - reject for two (2) or more bond failures.
2/ Certification of conformance required (see 6.2).
3/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Microcircuit, Linear, Voltage Comparators				9292967
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire Bond strength	5	1/	3.2	Pull Tester
102	Wire Material	5	3/	3.2	Visual
103	Solderability	5	3/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics		2/	3.2	MIL-M-63323
105	Screening		2/	3.2	MIL-M-63323
106	External visual		2/	3.2	MIL-M-63323
107	Approved part/vendor verification	5	3/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
	Item requires special handling - See 4.5.18				

NOTES:
1/ Pull all wires on each device - reject for two (2) or more bond failures.
2/ Certification of conformance required (see 6.2).
3/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Microcircuit, Digital, CMOS (Initial Logic)				9292962
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire bond strength	5	1/	3.2	Pull Tester
102	Wire Material	5	4/	3.2	Visual
103	Solderability	5	4/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics 3/		2/	3.2	MIL-M-63320
105	Screening 3/		2/	3.2	MIL-M-63320
106	External visual 3/		2/	3.2	MIL-M-63320
107	Approved part/vendor verification	5	4/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
	Item requires special handling - See 4.5.18				

NOTES: 1/ Pull all wires on each device - reject for two (2) or more bond failures. 2/ Certification of conformance required (see 6.2). 3/ Government source inspection required. 4/ Reject for one (1) or more defective.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-E-63300A(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Microcircuit, Digital, CMOS (Final Logic)				9292963
					NEXT HIGHER ASSEMBLY
					9292965
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Wire bond strength	5	1/	3.2	Pull Tester
102	Wire Material	5	4/	3.2	Visual
103	Solderability	5	4/	3.2	MIL-STD-883, Method 2003
104	Electrical characteristics 3/		2/	3.2	MIL-M-63321
105	Screening 3/		2/	3.2	MIL-M-63321
106	External visual 3/		2/	3.2	MIL-M-63321
107	Approved part/vendor verification	5	4/	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.9	Visual
	Item requires special handling - See 4.5.18				

NOTES: 1/ Pull all wires on each device - reject for two (2) or more bond failures. 2/ Certification of conformance required (see 6.2). 3/ Government source inspection required. 4/ Reject for one (1) or more defective

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4.4.3 Testing. Testing is described in the First Article and Quality Conformance Inspection "Classification of Defects and Test" forms and as specified in 4.5.

4.4.3.1 Ambient (25°C ± 5°C), Hot (63°C + 3°C) and Cold (-37°C - 3°C) temperature functioning tests. The table below shall be used for the selection of the lot acceptance sample sizes for ambient, hot and cold temperature testing of lots of electronic assemblies.

INSPECTION LOT SIZE	SAMPLE SIZE	* AMB	HOT	COLD	ACCEPT - REJECT CRITICAL/SPECIAL DEFECTS	MAJOR DEFECTS
2-500	32	12	10	10	0-1	0-1
501-3200	125	45	40	40	0-1	1-2
3201-10,000	200	68	66	66	0-1	2-3
10,001-35,000	315	105	105	105	0-1	3-4

The units shall be tested for all the functional parameters (under critical, special, major) listed in 4.4.2.5. Reject number is the number of defective units that would reject the lot. since all the units have to be 100% inspected at ambient for critical and special defects the quantities listed under * do not have to be reinspected for critical and special defects at ambient temperature (see 4.5.16) but must be inspected for major defects.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in Classification of Defects and Tests forms beginning with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and section 6.3 herein.

4.5 Test methods and procedures . Only government approved test equipment and test fixtures shall be connected to electronic assemblies under test to verify that it complies with the requirements specified herein. Increasing the PTB oscillator frequency to reduce test time is acceptable provided, the higher frequency oscillator signal maintains an approximate 50% duty cycle and has a peak amplitude frequency oscillator signal which does not cause or mask malfunctions in the functional operation of the electronic assembly; the higher frequency oscillator signal is well within the capability of the test equipment to monitor PA, TL, SD (normal or extended) with an accuracy of ± 3 PTB counts; the actual

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procedure used to increase the frequency of the signal is approved by the government before it is implemented. When the PTB oscillator frequency is increased, the TTB oscillator frequency is to be proportionately increased or the TTB oscillator inhibited as applicable for the test being implemented. All testing shall be performed, unless otherwise specified at a supply voltage of 6.50 ± 0.10 volts, an ambient temperature of 25° ± 5°C, and a relative humidity of 60 ± 20%.

4.5.1 Quiescent current test. With the electronic assembly under test connected to the test fixture; the tripline-sensor interface circuit test points TP22, TP23, TP24 and TP25 connected to the ground; the piston actuator terminal (E33), the pressure cartridge terminal (E32), and the detonator terminal (E30) each connected to the ground through a 10 ohm resistor, measure the quiescent current of the electronic assembly. Observation shall be made for compliance with the requirement specified in 3.4.1. This is a non-destructive test.

4.5.2 Primary Time Base (PTB) period test,- With the electronic assembly under test connected to the test fixture and with both the primary and test time base circuits oscillating at their natural frequency, measure the period of the PTB oscillator at test point TP32 . Observation shall be made for compliance with the applicable requirement, 3.4.2.1. This is a non-destructive test.

4.5.3 Test Time Base (TTB) period test.- With the electronic assembly under test connected to the test fixture and with both the primary and test time base oscillators operating at their natural frequency, measure the period of the TTB oscillator at test point TP35. Observation shall be made for compliance with the applicable requirement, 3.4.2.2. This is a non-destructive test.

4.5.4 S&A assembly arm signal test. With the electronic assembly under test connected to the test fixture, verify that the S&A assembly arm signal occurs at the PTB clock count specified in Table I of Drawing 9292960 (See 3.4.2.3). This is a non-destructive test.

4.5.5 Pressure cartridge signal test. With the electronic assembly under test connected to the test fixture, verify that the pressure cartridge signal occurs at the PTB clock count specified in Table I of Drawing 9292960 (See 3.4.2.4). This is a non-destructive test.

4.5.6 Electronic arm test. With the electronic assembly under test connected to the test fixture, verify that electronic arming

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occurs at the PTB clock count specified in Table I of Drawing 9292960 (See 3.4.2.5). Make the verification by incorporating, as a minimum, the following checks in the test procedure:

a. At the PTB clock count of 3 (±2) less than that required for electronic arm, open circuit one of the four tripline-sensor interface circuit input terminal and confirm that the detonator firing circuit does not turn on.

b. At the PTB clock count of 3 (±2) greater than that required for electronic arm, open circuit one of the four tripline-sensor interface circuit input terminal and confirm that the detonator firing circuit is turned on,

This is a non–destructive test.

4.5.7 Low voltage detect test. With the electronic assembly under test connected to the test fixture, verify that the low voltage detect circuit initiates the turn on of the detonator firing circuit when the supply voltage is reduced to the low voltage detect circuit trigger level specified in TABLE II of Drawing 9292960 (See 3.4.2.6). For this test it is required that both the PTB and TTB oscillators be operating at their natural frequency, and that the initial supply voltage be 6,5 ± 0.10 volts and linearly reduced no faster than 50 millivolts/125 milliseconds until the low voltage detect circuit is triggered. Observation shall be made for compliance with the applicable requirement. This is a non-destructive test.

4.5.8 Extended self-destruct signal test. With the electronic assembly under test connected to the test fixture and the fuses simulated open by connecting test point pad TP6 to the positive voltage supply, verify that the extended self-destruct signals, that initiate the turn on of the detonator firing circuit, occur at pins 1 and 15 of U2 at the PTB clock count specified in TABLE I of Drawing 9292960 (See 3.4.2.7). This test is to be conducted before the normal self–destruct test (4.5.9). This is a non-destructive test.

4.5.9 Normal self-destruct signal. With the electronic assembly under test connected to the test fixture, verify that the normal self-destruct signals, that initiate the turn of the detonator firing circuit, occur at pins 1 and 15 of U2 at the PTB clock count specified in Table I of Drawing 9292960 (See 3.4.2.8). This is an non-destructive test.

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4.5.10 Antidisturbance function mode. With the electronic assembly under test connected to the test fixture so that its plane makes an approximate 45 degree angle with the horizontal, verify the ant i-disturbance function mode by slowly rotating the electronic assembly about the cylindrical axis and confirming that the detonator firing circuit turns on before a 360 degree rotation is made (See 3.4.3.1). This is a non-destructive test.

a. Alternate procedure. With the electronic assembly under test connected to the test fixture so that its plane is parallel with the plane of the test fixture, verify the antidisturbance function made by passing a permanent magnet across the top of the AD switch assembly. Proper function will be evidenced by confirming the resulting turn-on of the detonator firing circuit. The magnet may make brief physical contact to the top of the AD switch assembly during this motion. The magnet flux density shall be no less than 20 gauss and no greater than 60 gauss. The magnet holding force shall be 0.7 pounds (minimum).

4.5.11 Tripline sensitivity test. With the electronic assembly under test connected to the test fixture, verify that each input of the tripline-sensor interface circuit meets the sensitivity requirement specified in 3.4.3.2. Make this verification by incorporating, as a minimum, the following checks in the test procedures:

a. Prior to the electronic assembly power up, check and verify that the resistance at each tripline-sensor interface circuit input (TP-22, TP-23, TP-24, and TP-25) with respect to ground is less than 10 ohms.

b. After electronic arm and with the resistance at the tripline sensor interface input under test at less than 10 ohms, confirm that the detonator firing circuit is not turned on when this resistance is increased to 3.9K ohms. Confirm that the detonator firing circuit is turned on when the resistance is increased from 3.9K to 10K ohms. The change in resistance from 10 ohms to 3.9 ohms and from 3.9K to 10K ohms shall be made such that no open circuit condition exists during the change.

c. Reset the electronic assembly and repeat (b) above for each tripline-sensor interface circuit input. Non-destructive test.

4.5.12 PTB oscillator circuit check. With the electronic assembly under test connected to the test fixture and both the PTB and TTB oscillator circuits operating at their natural frequencies, verify that the oscillator check circuit is operational by

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confirming that the oscillator check circuit initiates the turn on of the detonator firing circuit within 32 TTB clock counts after the PTB oscillator is stopped (See 3.4.4). This is a non-destructive test.

4.5.13 Firing circuit energy. With the electronic assembly under test connected to the test fixture, verify that the piston actuator, pressure cartridge, and detonator firing circuits meet the requirements specified in 3.4.5, 3.4.5.1, 3.4.5.2 and 3.4.5.3. This is a non-destructive test.

4.5.14 Self-sterilization function, With the electronic assembly under test connected to the test fixture and the detonator firing circuit output E30 an open circuit, verify that the voltage of the firing capacitor is less than 1.3 volts 5 seconds after the firing circuit is turned on (See 3.4.6). This is a non-destructive test.

4.5.15 Reset circuit. With a linear increasing voltage (zero to 6.5 ± 0.10 d.c. volts in 50 milliseconds rein) applied to VDD of the electronic assembly from an external d.c. power supply, verify that at reset the output "PA" at test TP 20 of the initial logic is low and the output "D" or "F" at TP 17 of the final logic is low. Observation shall be made for compliance with requirements of 3.4.7. Non-destructive test.

4.5.16 Ambient, hot and cold temperature functional testing. The sample of electronic assemblies specified in 4,4.3.1 shall be subjected to all functional tests at the operating temperature extremes. The units shall be conditioned at the specified temperatures (see 4.4.3.1) for two hours prior to the test. Testing shall be performed with the unit in the temperature chamber. Approved functional test equipment shall be used. Acceptable threshold levels are specified on DWG 9292960 for the temperature extremes. Acceptance limits for the digital circuitry (SD timing control) at the temperature extremes are specified on DWG 9292960. Observation shall be made for compliance with 3.5 and 3.6. Non-destructive test.

4.5.17 Printed wiring board cleaning procedures. Printed wiring board assembly cleaning procedures shall be documented and supplied to the Government (See 6.5.) Verification of adherence to the cleaning procedure shall be done on a daily basis (Ref. Data Item DI-T-5372 (tailored)).

4.5.18 Integrated circuit handling procedures. Integrated circuits require care and handling to avoid damage due to static discharge. The contractor shall document his handling procedure for the integrated circuits and supply the procedure to the government (See 6.5.) Adherence to the handling procedure shall be verified daily. (Ref. DI-T-5372 (tailored)).

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4.5.19 Quality assurance soldering plan. The QA soldering plan as noted in 3.3.2 shall be submitted to the Government. (see 6.5). (Ref. DI-T-5372 (tailored)).

4.5.20 Data requirements. The contractor shall submit a quality inspection test, demonstration and evaluation report in accordance with Data Item Description DI-R-1724 (tailored). It shall include all detailed inspection and test results. Test data reports shall be submitted to the technical agency listed in 6.5.

4.5.21 Variables data. The contractor shall submit variables data in accordance with Data Item Description DI-R-1724 (tailored) for the first article samples. It shall include all detailed inspection and test results with respect to the following parameters: Primary Time Oscillator Period, Test Time Base Oscillator Period, Quiescent Current, Low Voltage Detection level, TL Sensor Threshold level. Test data reports shall be submitted to ARRADCOM (6.5).

4.5.22 Switch assembly (Anti-disturbance switch).

4.5.22.1 AD switch function and closure time duration. The switch assembly shall be placed in an approved fixture. The fixture shall be capable of rotating the switch assembly through 360° on its cylindrical axis, while the axis is held at 15° ± 2° referenced to the horizontal, for both CW and CCW directions of rotation. Verify that the functional requirements of the drawing are met. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.22.2 Insulation resistance of AD switch. The switch assembly when in a suitable fixture shall be measured for insulation resistance in the reset position using an approved megohmmeter. Verify that the switch assembly does meet the requirements of the drawing. This is a non-destructive test and acceptable parts may be returned to the lot.

4.5.22.3 Plating of AD switch can. Five (5) cans shall be randomly selected from each plating batch. The specimens shall be cross-sectioned through the axial center of the can, polished and examined microscopically to measure the plating thickness. If the total average plating thickness for the five cans fails to meet the requirements as specified on the drawing, the sample shall be classed defective and the lot shall be rejected. (Destructive test).

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging

5. 1.1 Level A. The Electronics Assembly shall be preserved and packaged in accordance with the requirements of MIL-E-17555 as applicable to printed circuit boards. Each electronic assembly shall be individually cushioned and secured to prevent any movement or damage to any of its parts. Folding of the flexible PWC leads is not permitted.

5.2 Packing.

5.2.1 Level A. The Electronics Assembly, preserved and packaged as above, shall be packed in accordance with the requirements of MIL-E-17555 as applicable to printed circuit boards.

5.3 Marking. Marking of the outer box shall comply with the commerical requirements of MIL-STD-1188. Additional marking indicating the items to be fragile electronics shall be employed.

5.4 Shipping. When electronic assembly packages or more than one lot are shipped together, each lot shall be kept separate and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Intended use. The electronics assembly covered by this specification is intended for use as a low power timing base and function generator for use in Mine, Antipersonnel: HE, M74.

6.2 Ordering data. Procurement documents shall specify the following:

a. Title, number, and date of this specification.

b. Provisions for submission of first article samples.

c. Provisions of MIL-A-48078.

d. Certification of conformance.

6.3 Submission of inspection equipment designs for approval. (See MIL-A-48078) Submit equipment designs as required to: Commander, U.S Army Armament Research and Development Command, ATTN: DRDAR-QAA-1, Dover, New Jersey 07801.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Command (ARIMDCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of A RRADCOM.

6.5 Inspection and test results, process controls. Inspection, and test results inspections and process controls procedures shall be sent to: Commander, U.S Army Armament Research and Development Command Attn: DRDAR-QAA-M, Dover, New Jersey 07801.

6.6 Data requirements. Deliverable data required by this specification is cited in the following paragraph:

Paragraph	DATA Requirements	Applicable DID
4.5.17	Printed wiring board cleaning procedures	DI-T-5372 (Tailored)
4.5.18	Integrated circuit handling procedure	DI-T-5372 (Tailored)
4.5.19	Quality assurance soldering plan	DI-T-5372 (Tailored)
3.7, 3.7.1	Quality Inspection Test	
4.4.2.3	Demonstration and Evaluation	DI-R-1724
4.5.20, 4.5.21	Report	(Tailored)

6.7 Special defect. A defect, other than critical (see MIL-A-48078) that judgement and experience indicate may, depending upon the degree of variance from the design requirement:

a. Result in hazardous or unsafe conditions for individuals using, maintaining or depending upon the product or

b. Prevent performance of the tactical function of a major end item.

6.8 Change from previous issue. Due to the extensiveness of the changes, asterisks to identify changes with respect to the previous issue are not used in this revision.

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Project No: 1345-A163

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APPENDIX

SUPPLEMENTAL SOLDERING PROVISIONS

10 SCOPE: This appendix details supplemental soldering provisions MIL-S-45743 for hand soldering and to MIL-S-46844 for machine soldering.

20 APPLICABLE DOCUMENTS: See 2 herein.

30 EXCEPTIONS AND CLARIFICATIONS:

30.1 Exceptions/clarifications to MIL-S-45743. The following exceptions/clarifications to reference MIL-S-45743 paragraph titles shall apply.

Control of airborne contaminants. Delete the sentence, "Positive pressure shall ..... between them."

Soldering irons and resistance soldering irons. Delete the sentence, "The soldering iron shall . . . ..or standby condition.", and substitute the sentence, "The soldering iron shall be a soldering iron approved by the technical agency or a soldering iron that attains a temperature no more than 10°F (5.6°C) above the control temperature when in an idling or standby condition."

Printed board terminations., Subparagraph 'b'. Change to read, ".... is permissible. Leads shall not extend, beyond the terminal pad or printed wiring pattern, far enough to cause less than .015 inches of clearance between conductors."

Printed board terminations., Subparagraph 'd'. Change to read, "When leads cannot be clinched per above, they shall extend through the printed board, be visible and properly soldered."

Ribbon lead geometry. Add new sentence, "Ribbon lead termination from battery and lead assembly, shall comply with "Printed board terminations", subparagraph 'b'."

Cleaning the soldered connection. Change to read, ... (within one (1) hour) using solvents or combination of solvents which will remove residual flux, activators, and other contaminants."

Resistivity of solvent extract. Delete.

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One hundred percent inspection. Delete and substitute the following: "Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table II of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

Sampling inspection. Delete.

Visual inspection. Change to read, ".... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

Cleanliness test. Delete.

30.2 Exceptions/clarifications to MIL-S-46844. The following exceptions/clarifications to reference MIL-S-46844 paragraph titles shall apply.

Flux application. Change to read . . . . . even coated surface. Any thinner used shall be a product from the flux manufacturer recommended for the specified flux being utilized or a substitute approved by the technical agency."

The solder contact. Add: "Bonded components or clinched leads are acceptable restraints provided they are implemented in a manner to assure that the leads do not move during the solder cycle."

Cleaning. Change to read . . . . . scraping shall not be used. Residual flux, activators and soils shall be removed to the degree approved by the technical agency using solvents or combination of solvents which will remove contaminants."

Resistivity of solvent extract. Delete.

One hundred percent inspection. Delete and substitute the following: "Level of inspection. Inspection of soldered connections and associated wiring and leads shall be performed to the level specified in the classification of defects in the product specification. Table 11 of MIL-S-45743 provides a list of defects with reference to the applicable requirements."

Sampling inspection, Delete.

visual inspection Change to read, ... a clear incandescent or fluorescent light should be used for detection of uncovered copper."

Cleanliness test. Delete.

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40 SUPPLEMENTAL SOLDER WORKMANSHIP STANDARDS

40.1 General. The following solder workmanship standards supplement the workmanship requirements contained in MIL-S-45743 and MIL-S-46844 and are provided as additional aids in establishing the minimum acceptance standards for soldering. All examinations will be performed using a minimum 5 diopter lens. Suspected connections may be examined using 10x magnification to determine acceptability.

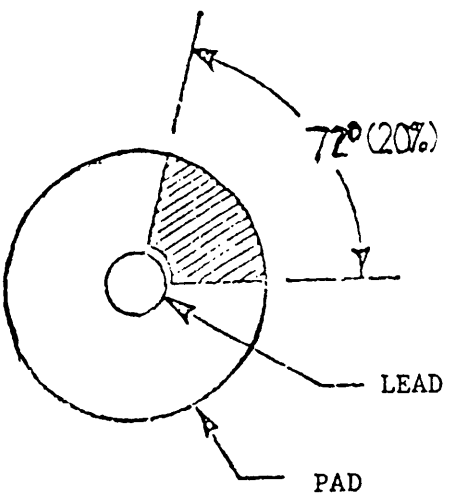
40.2 Definitions.

40.2.1 Voids. All holes, pits, and voids regardless of size are classified as voids.

40.2.1.2 Hole with solder plug. The total accumulative void area on each side of hole is less than 20% of the hole area and does not expose base metal. Also there are no voids through the board. Evidence of good wetting at the pad surface,

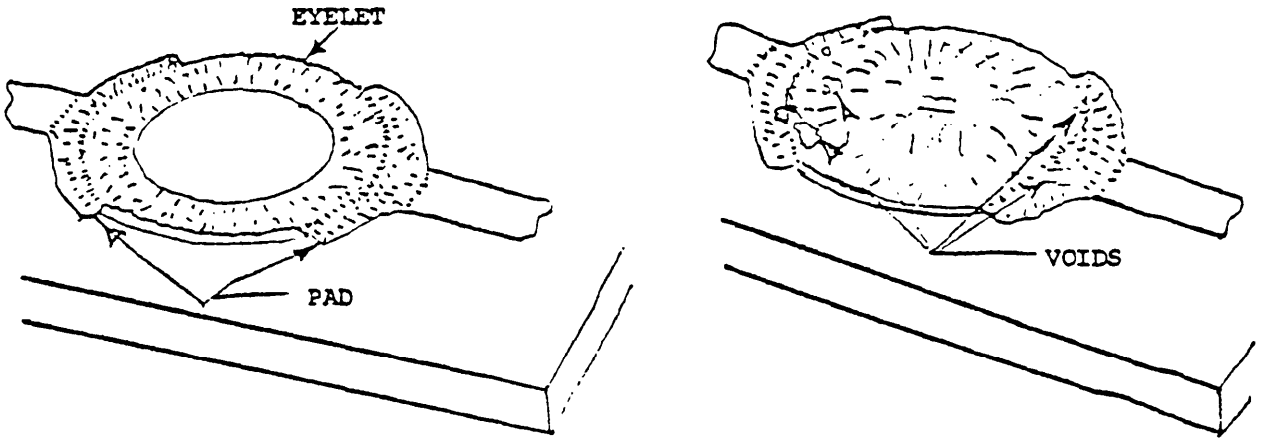
40.2.1.3 Hole with leadwire. For solder side and component side: the total accumulative void area is less than 20% of the lead circumference; does not visibly expose base metal; leads adjacent to voids exhibit good wetting and 100% solder coverage of the lead surface. There are no voids through the board. The lead and pad are wetted.

40.2.1.4 Twenty (20) % void. (For solder joint with a lead.) Total void area equivalent to an area subtended by an arc equal to 20% of the lead circumference.



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40.2.1.5 Eyelet connection voids. The total cumulative area of all voids is less than 20% of the total interface between eyelet and pad. Void(s) do not visibly expose base metal. The presence or lack of solder in the eyelet hole is not basis for nonacceptability.

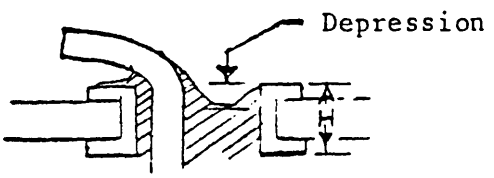


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40.2.2 Maximum solder. A solder joint where it can be established that the lead exists and solder does not overflow beyond the confines of the connection area (the pad and tracks). Lead and pad must be well wetted.

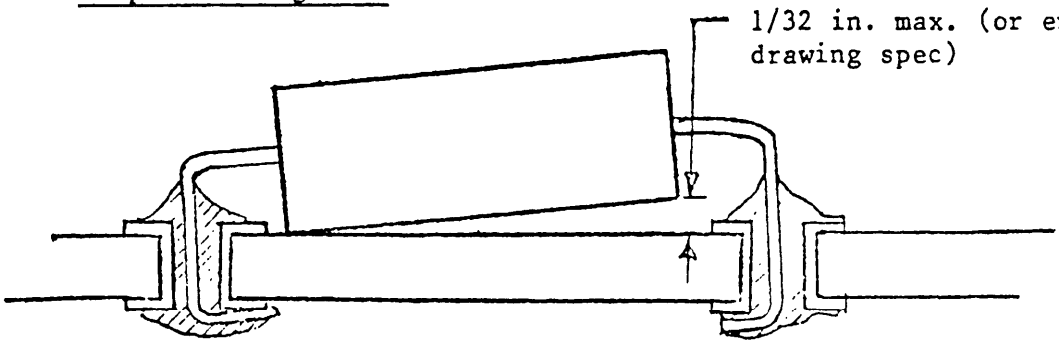
40.3 Lifting of pad. Lifted pads shall be controlled. Quantity of boards with lifted pads shall be controlled by the contractor as follows: Lifted pads exceeding 1/3 lifted to 0.40% AQL. Lifted pads less than 1/3 lifted is 1.0% AQL. Damage must not be in quadrant enclosing track. Damage not to break into plated through hole. No damage to opposite hole.

40.4 Lead, solder side. Exposed end (except copper) are visible at the end of the end. Lead is not closer than .015" to adjoining conductive circuitry. Maximum solder is permissible. Solder fillet is depressed, but the depression is less than 25% of the board thickness.



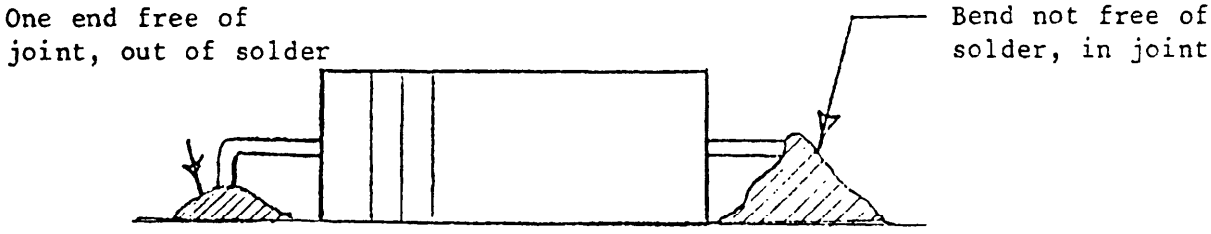
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40.5 Component alignment.

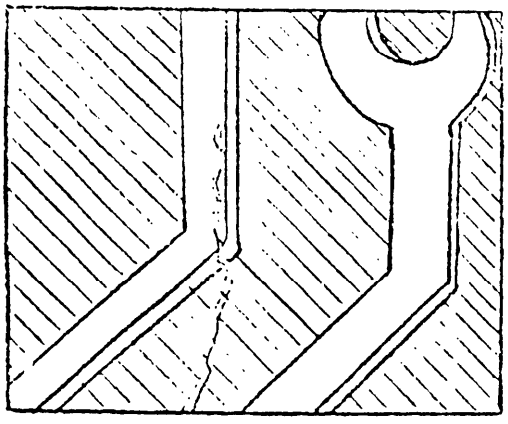


Vertical misalignment of component and printed wiring board is not in excess of 1/32 in. (or as specified on engineering drawing).

40.6 Strain relief. For axial lead components having a 0.080 max. diameter body allow one lead bend to be in solder joint, providing opposite lead bend is free of solder.



40.7 Scratches.



1. Scratch does not reduce cross-sectional area of conductor more than 20%

2. Does not visibly expose base metal.

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50 PRINTED WIRING BOARD ASSEMBLIES

Following wave soldering, printed wiring board assemblies shall be inspected (under adequate lighting and magnification) for soldering workmanship.

50.1 Major Defect Inspection

a. Inspection shall be to a 0.40 percent AQL, MIL-STD-105, Level II, normal sampling plan.

b. A board will be defined as defective during inspection if:

1. The bottom contains any joints defined as unacceptable by any of the documentation in sections 20, 30 and 40 of this appendix.

2. The top contains any joint defined as unacceptable by the documentation listed in 1 above except twelve (12) joints are allowed to be dewetted, A dewetted joint on the top must have evidence of solder in the hole when viewed from the top.

3. A maximum of thirty- five (35) joints with O to 20% voids will be allowed on each board; and no more than twenty (20) joints with 0-20% voids will be allowed on any one side of the printed wiring board.

50.2 Minor Defect Inspection

a. Inspection shall be on the sample size as specified in paragraph 50.1 above.

b. A board will be defined as defective if it contains three (3) or more minor defects.

c. Accept/reject criteria for the lot shall be to a 4.0% AQL, MIL-STD-105, Level II, normal sampling plan.

d. Minor defects are defined as the following:

1. Conductive surfaces closer than .010″ but greater than .005″ assembly characteristics only, not relative to the printed wire board construction.

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2. Copper showing -

(a) Clipped lead - evidence of good wetting and reflow must be present.

(b) Non-clipped lead - characteristic of good solder joint, i.e., wetting, etc., must be present.

3. Solder splash (attached and not shorted).

4. Solder balls less than .005″.

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NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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